



TQIS LTD

ENAGEED RESOURCE

WELL NAME: OKI_OZIENGBE SOUTH-5

RIG NAME: HPEB 187

FIELD/STATE: OKI OZIENGBE-EDO STATE



MUD LOG (MD)

@ 1:1000ft

LEGEND

WOB - Weight on Bit
BG - Background Gas
CB - Core Bit
CG - Connection Gas
CK - Filter Cake
CKF - Check For Flow
CL - Salinity ppm
CO - Circulate Out
CR - Circulate Returns
CT - Carbide Test
DB - Diamond Bit
DCB - Diamond Corebit
DC - Depth Correction

DF - Derrick Floor
DG - Drilling Gas
DS - Directional Survey
DST - Drill Stem Test
EL - Electric Log
EMA - Equivalent c2 in Air
F - Filtrate API cc's
F/T - Flowline Temperature
GCM - Gas Cut Mud
GCW - Gas Cut Water
GOCM - Gas and Oil Cut Mud
GL - Ground Level
KB - Kelly Bushing

LAT - Log After Trip
LC - Lost Circulation
NB - New Bit
NCB - New Core Bit
NR - No Returns
PDCB - Polycrystalline Diam.
Compact Bit
PPM - Parts Per Million
PR - Poor Returns
SPP - Pump Pressure
PV - Plastic Viscosity
RM - Mud Resistivity
RMC - Mud Cake Resistivity

PEFR - Perforated
RMF - Mud Filtrate Resistivity
RPM - Revolution/Minute
RRB - Return Bit
RT - Rotary Table
SD - Sand%
SF - Sea Floor
SO - Show of Oil
SOL - Solids%
STG - Short Trip Gas
SWG - Swab Gas
SVG - Survey Gas
S/T - Suction Temp.

TB - Turbo Drill
TD - Total Depth
TG - Trip Gas
TCL - Trip Chlorides
TVD - True Vert. Depth
V - Viscosity API seconds
W - Weight of Mud ppg
GR - Gamma Ray

Limestone



Dolomite



Siltstone



Sandstone



Sand



Lignite



Cement



Shale



Clay



Spud Date: 30/01/2014
Logging Started: 30/01/2014
Logging Finished: 15/03/2014
Start Depth: 500ft
Finished Depth: 12873ft

ELEVATION

Permnt Datum:
Kelly Bushing: N/A
Water Depth: N/A
Ground Level: 51.64ft

LOCATION

Lease: OPL 274
Geog: N 6 3 40.66" ; E 5 25 24.99"
Colony: 332963.56mE ; 227953.22mN

BOREHOLE RECORD

CASING RECORD

Size	From	To	Size	Weight	From	To
16"	0	6110'	26"	N/A	0	250'
12.25"	6110	9003'	13..375	68	0	6110'
8.5'	9003	12,873"	9.625"	47	0	9003'
			7"	29	9511'	11782'

Logging Personnels:

ALI .N, OTABOR . L, OBASEKI N, UGAH R,
EMMANUEL A, UTULU C. JAMES A, ATAMA Y, MIKE

Company Reps:

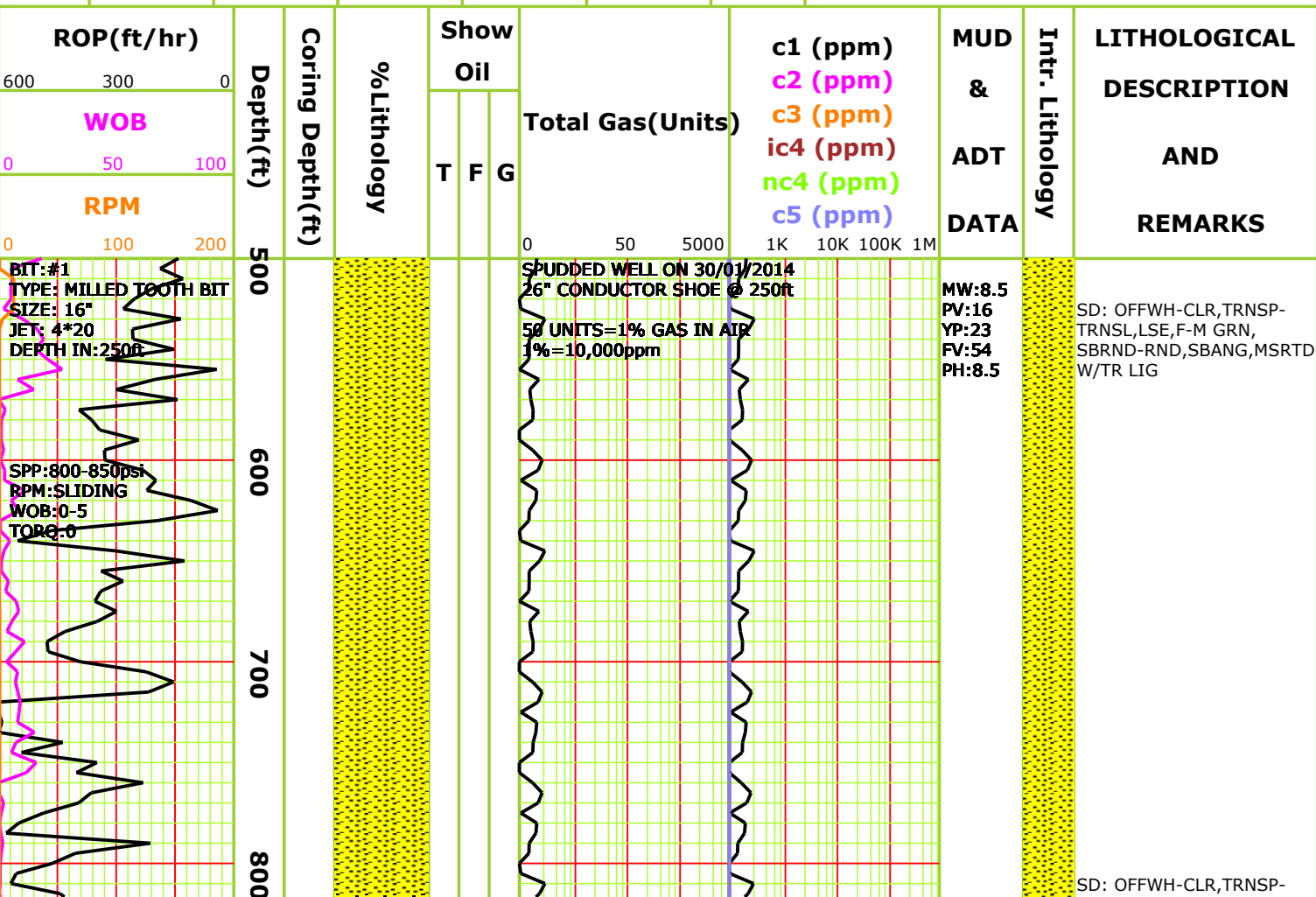
OSAGIE O. DEBOLA S. ANTHONY .N, IFEANYI N,
FRANCIS A

Wellsite Geologists:

BANKOLE. O, LANRE. O

Remarks:

OKI-OZIENGBE-SOUTH 5 WAS DRILLED
SUCCESSFULLY



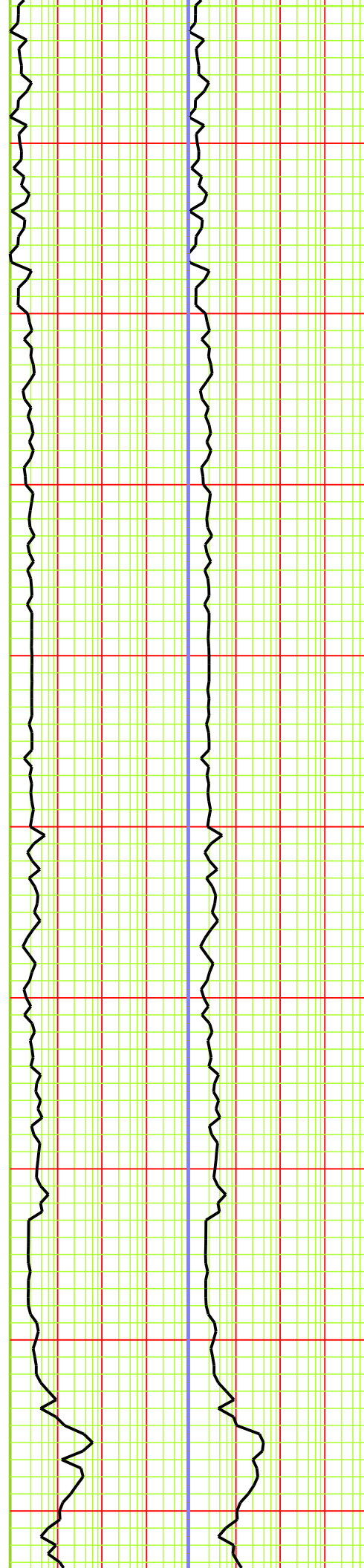
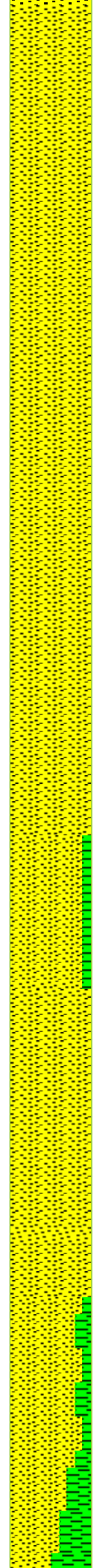
SD: OFFFWH-CLR, TRNSP-



MD:984.50ft
INCL:7.47
AZI:306.74
TVD:982.64ft

SPP:1200psi
RPM:40
WOB:0-15
TORQ:4-6

SPP:1350psi
RPM:40
WOB:0-10
TORQ:4-7



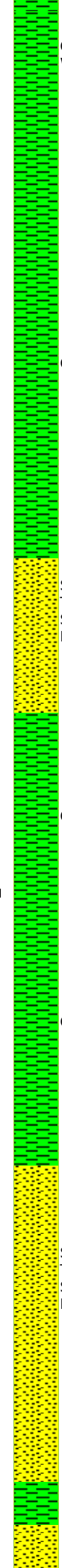
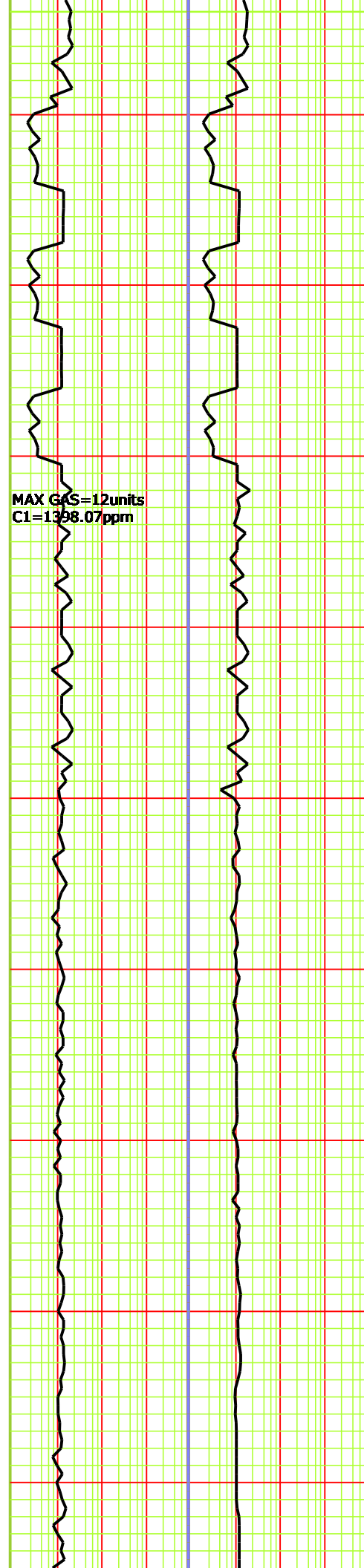
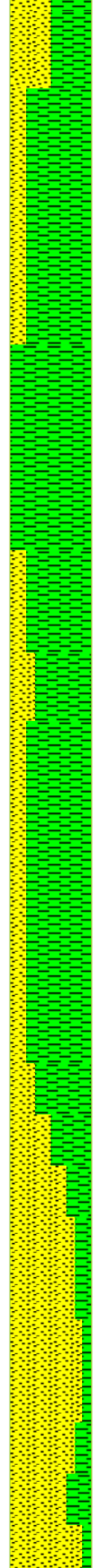
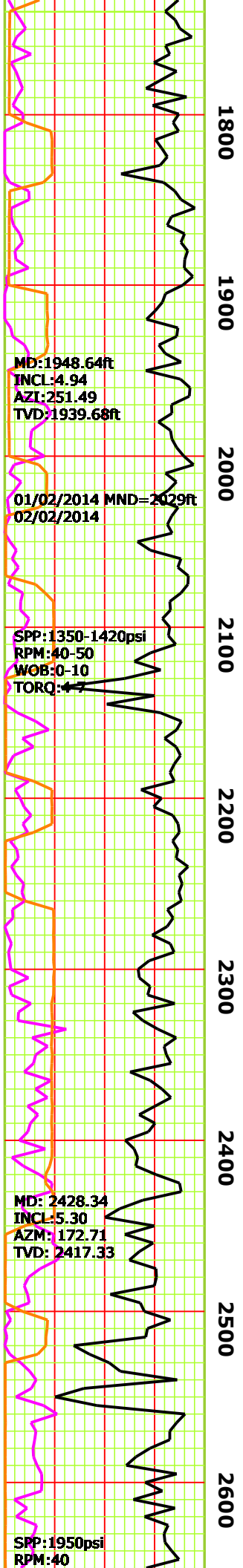
OCC C GRN,SBRND-SBANG,
MSRTD

SD:OFFWH-CLR,OCC LTBRN
TRNSP-TRNSL,UNCONS,M-C
GRN,SBRND-SBANG,PSRTD,
QTZ GRN,NO FLOR

SD:OFFWH-CLR,OCC LTBRN
TRNSP-TRNSL,F-M GRN,
SBRND-RND,SBANG,MSRTD,
NO FLOR

MW:8.5
PV: 16
YP: 23
FV:54
PH: 8.5

MW:8.5
PV:16
YP:23



CLST: LTGY, SFT, STKY,
WASHABLE

CLST: LTGY, SFT, STKY

SD: OFFWH-CLR, OCC LTBRN
TRNSP-TRNSL, F-M GRN,
SBRND-RND, SBANG, MSRTD,
NO FLOR

CLST: DKGY, SFT, STKY

CLST: DKGY, SFT, STKY

SD: OFFWH-CLR, OCC LTBRN
TRNSP-TRNSL, F-M GRN,
SBRND-RND, SBANG, MSRTD,
NO FLOR

WOB:0-10
TORQ:5-9

MD: 3007ft
INC: 12.22
AZM:174.93
TVD: 2987.74

02/02/2014 MND=3288ft
03/02/2014

SPP:1970
WOB:0-20
RPM:0-50
TOR: 0-7

03/02/2014 MND=3500ft
04/02/2014

2700
2800
2900
3000
3100
3200
3300
3400
3500

MAX GAS=12unit
C1=2361.46

MAX GAS=11units
C1=2181.14

MW:9.1
PV: 16
YP: 25
PH: 9.0
FV:70

MW: 9.2ppg
PV: 16
YP:28
PH: 9.0
FV: 72

SD:OFFWH-CLR,OCC LTBRN
TRNSP-TRNSL,F-M GRN,
SBRND-RND,SBANG,MSRTD,
NO FLOR

CLST: DKG,Y,SFT,STKY

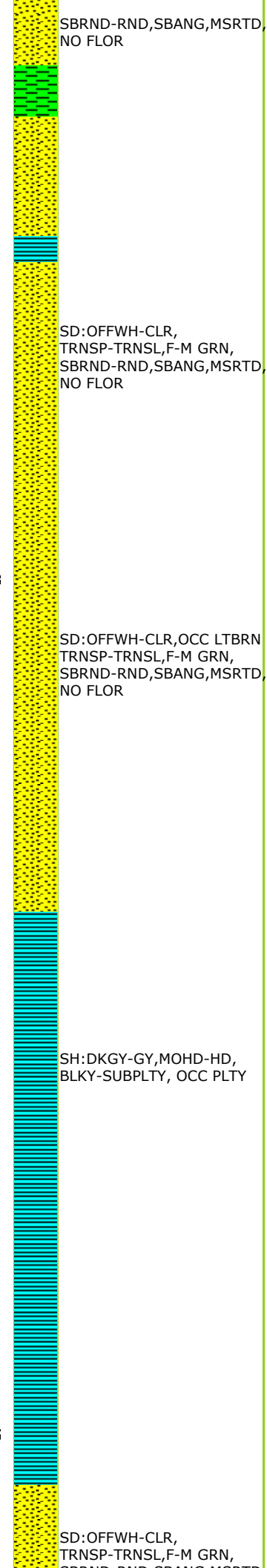
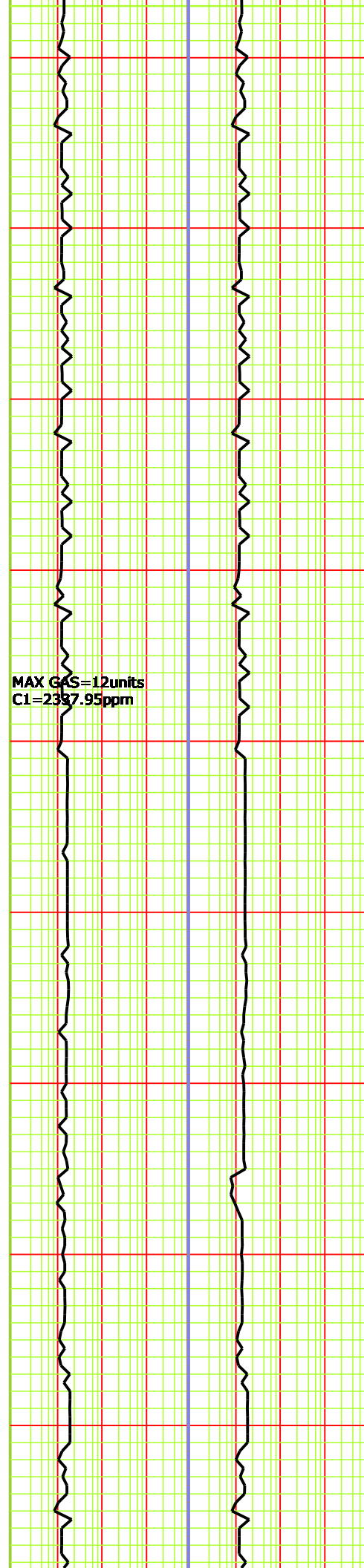
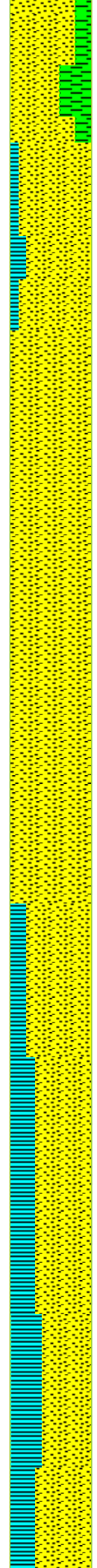
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TRNSP-TRNSL,F-M GRN,
SBRND-RND,SBANG,MSRTD,
NO FLOR

CLST: DKG,Y,SFT,STKY

SD:OFFWH-CLR,
TRNSP-TRNSL,F-M GRN,
SBRND-RND,SBANG,MSRTD,
NO FLOR

CLST: DKG,Y,SFT,STKY
WASHABLE

SD:OFFWH-CLR,



SBRND-RND,SBANG,MSRTD,
NO FLOR

SD:OFFWH-CLR,
TRNSP-TRNSL,F-M GRN,
SBRND-RND,SBANG,MSRTD,
NO FLOR

MW:9.3
PV:16
YP:28
PH:9
FV:72

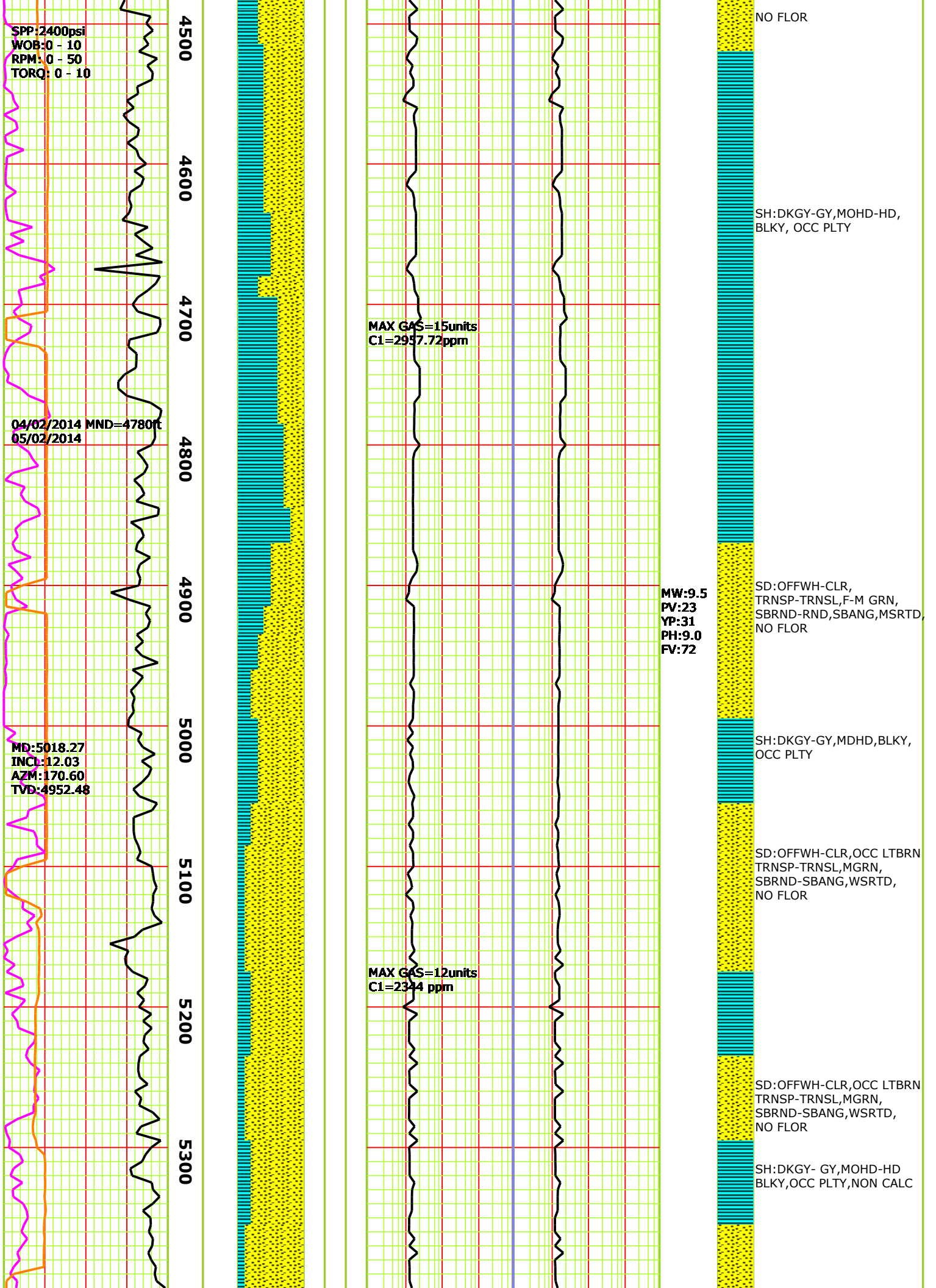
SD:OFFWH-CLR,OCC LTBRN
TRNSP-TRNSL,F-M GRN,
SBRND-RND,SBANG,MSRTD,
NO FLOR

MAX GAS=12units
C1=2387.95ppm

SH:DKGY-GY,MOHD-HD,
BLKY-SUBPLTY, OCC PLTY

MW:9.5
PV:16
YP:28
PH:9
FV:72

SD:OFFWH-CLR,
TRNSP-TRNSL,F-M GRN,
SBRND-RND,SBANG,MSRTD,
NO FLOR



SPP:2760
RPM:0 - 50
TORQ: 0 - 12
WOB: 0 - 10

400

5500

5600

5700

5800

5900

6000

6100

6200

6300

05/02/2014 MND=57156
06/02/2014

SPP:2890
RPM:0 - 55
TORQ:10-14
WOB:5 - 20

MD:5880.89
INCL:12.65
AZM:174.05
TVD:5796.58

BIT:#2
TYPE: PDC BIT
SIZE: 12.25"
JET: 5*15, 2*14

MAX GAS=10units
C1=1958.02 ppm

MAX GAS= 11 units
C1= 2127.83 ppm

MAX GAS=12 units
C1= 2157.79 ppm

MW:9.6
PV:23
YP:31
PH:9.0
FV:72

MW:9.6
PV: 23
YP: 31
PH: 8.5
FV: 72

SD:OFFWH-CLR,
TRNSP-TRNSL,F-MGRN,
SBRND-RND,SBANG,MSRTD-
WSRTD,NO FLOR

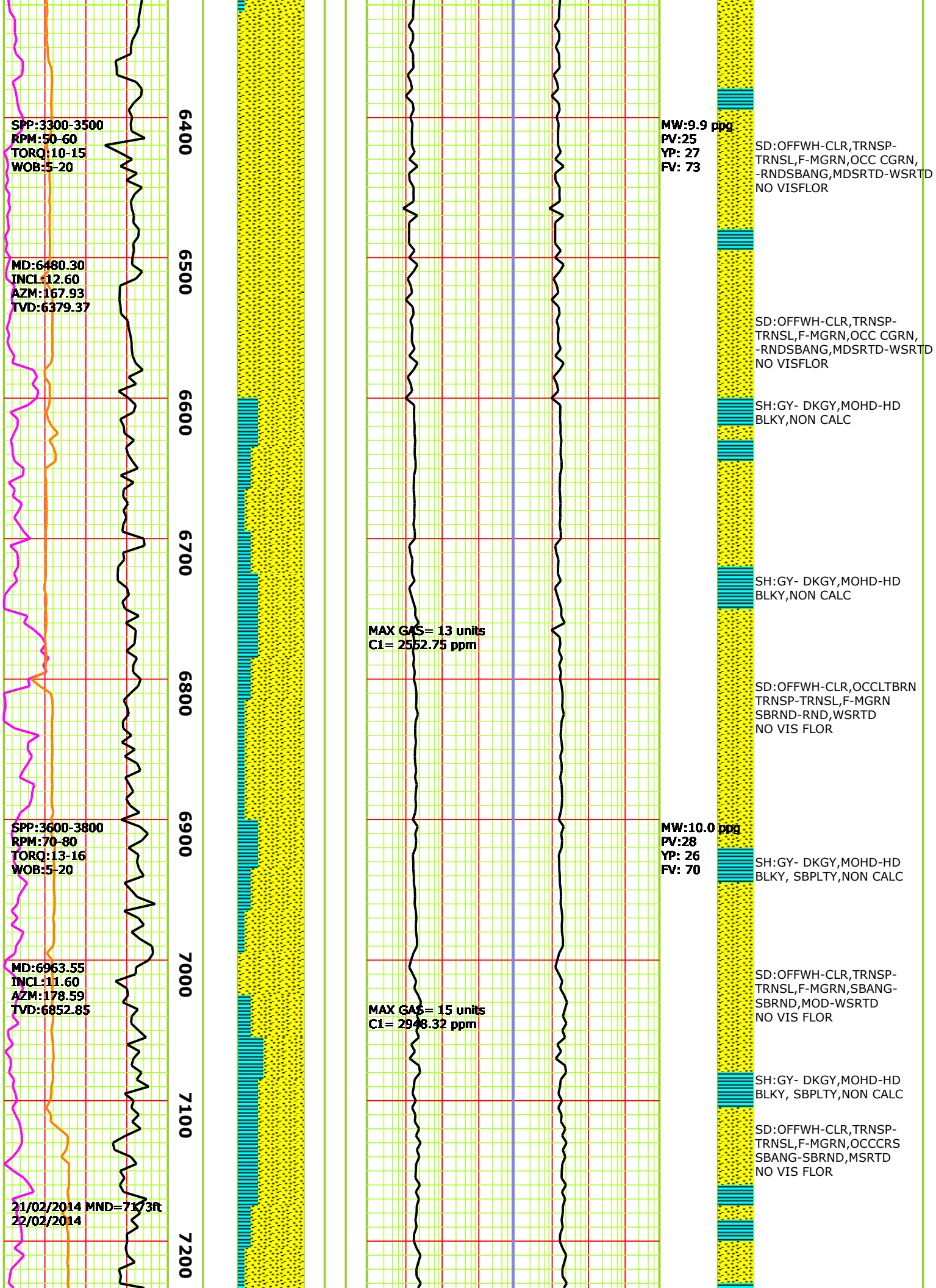
SD:OFFWH-CLR,
TRNSP-TRNSL,F-MGRN,
SBRND-RND,SBANG,MSRTD-
WSRTD,NO FLOR

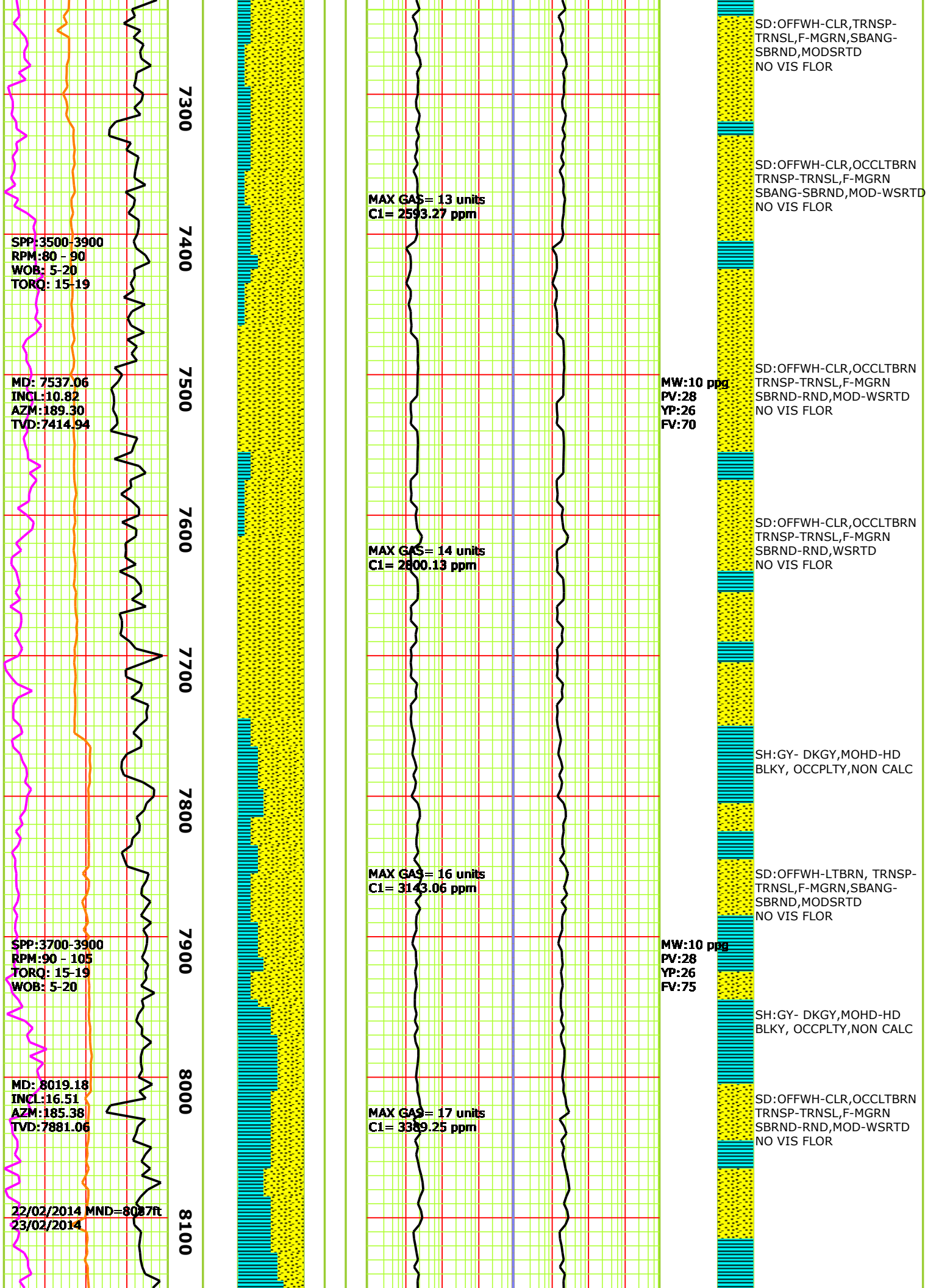
SD:OFFWH-CLR,
TRNSP-TRNSL,F-MGRN,
SBRND-RND,SBANG,MSRTD-
WSRTD,NO FLOR

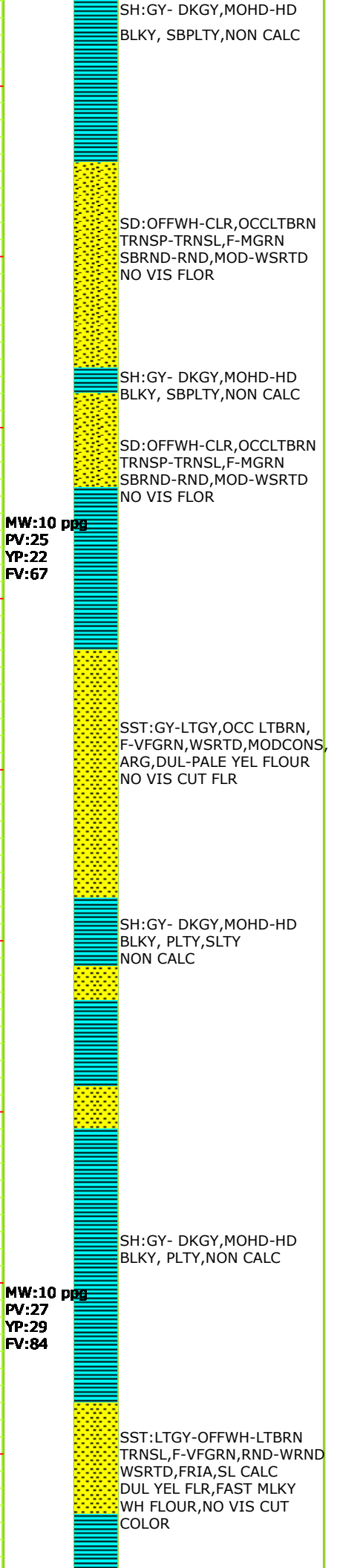
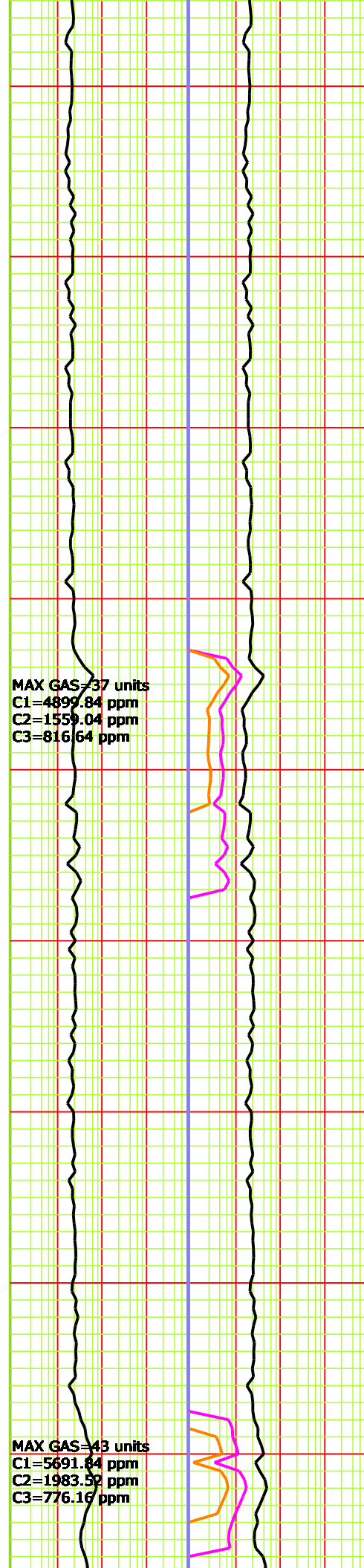
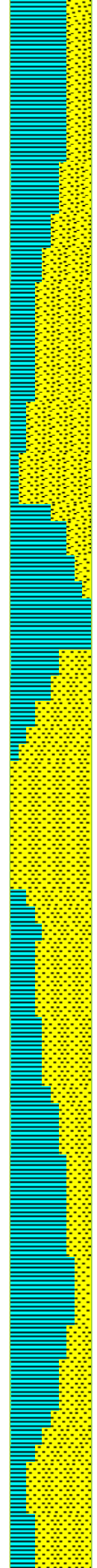
SD:OFFWH-CLR,TRNSP-
TRNSL,F-MGRN,SBRND-RND
SBANG,MDSRTD-WSRTD,NO
FLOR

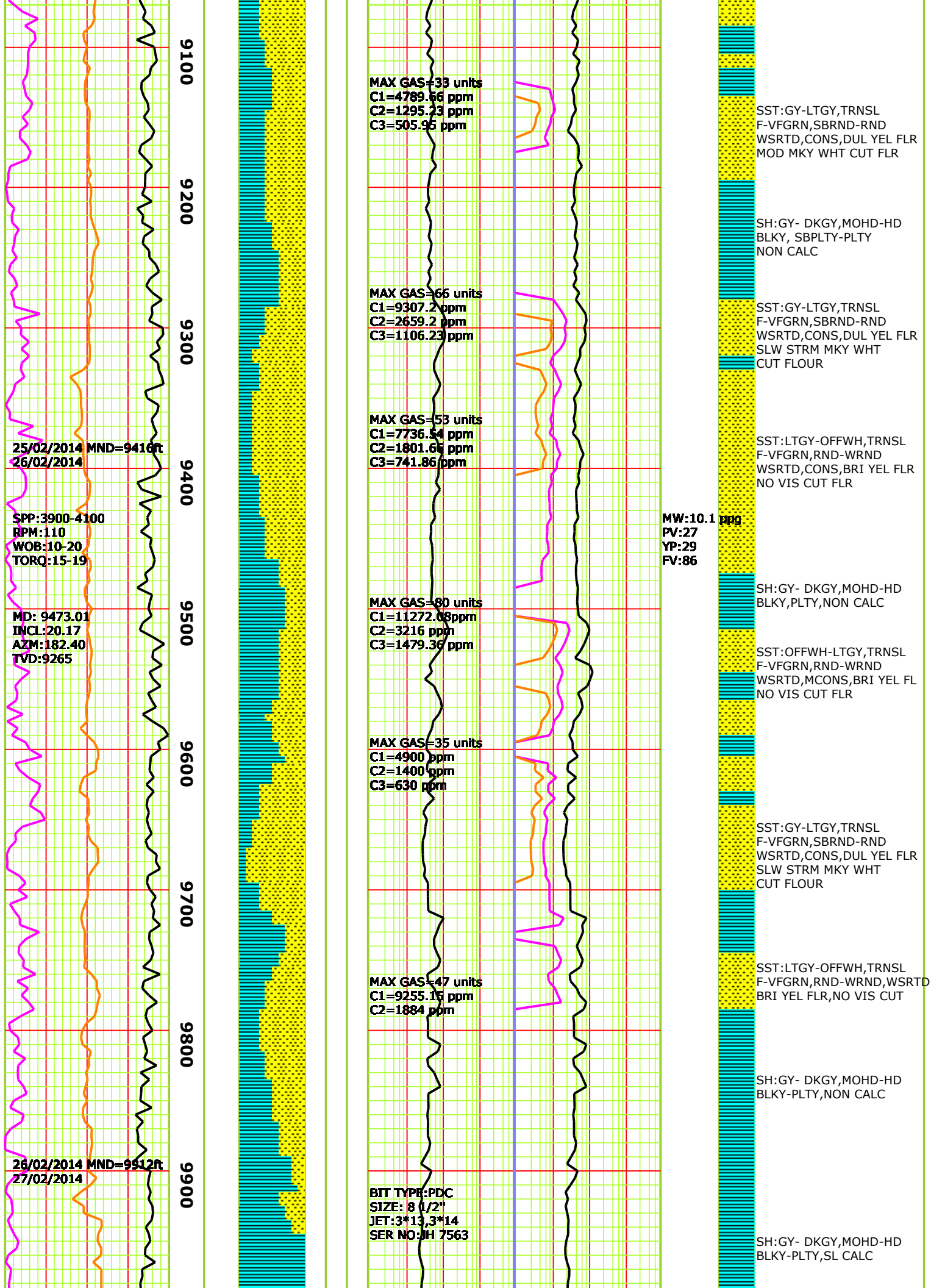
SH:DKGY- GY,MOHD-HD
BLKY,OCC PLTY,NON CALC

SD:OFFWH-CLR,TRNSP-
TRNSL,F-MGRN,SBRND-RND
SBANG,MDSRTD-WSRTD,NO
FLOR









MD: 9995.73
INCL: 28.34
AZM: 179.73
TVD: 9823.00

10000

10100

10200

10300

10400

10500

10600

10700

10800

09/03/2014 MND=10318ft
10/03/2014

SPP: 2700-3020
RPM: 110-120
WOB: 5-15
TORQ: 16-18

MD: 10668.93
INCL: 48.72
AZM: 180.33
TVD: 10243.61

MAX GAS=45 units
C1=6445.1 ppm
C2=1851.94 ppm
C3=760.94 ppm

MAX GAS=68 units
C1=7609.2
C2=2663.72
C3=1395.02
IC4=760.92

MAX GAS=49 units
C1=5997.6
C2=2097.16
C3=1099.56
IC4=599.76

MW: 10.2 ppg
PV: 27
YP: 31
FV: 80

MW: 10.2 ppg
PV: 27
YP: 31
FV: 80

SST: OFFWH-LTBRN, TRNSL
F-VFGRN, RND-WRND
WSRTD, FRI, DUL YEL FLR
MOD BLOOM MKY-WH
CUT FLR

SH: GY- DKGY, MOHD-HD
BLKY-PLTY, SUBFISS
NON CALC

SH: GY- DKGY, MOHD-HD
BLKY-PLTY, SUBFISS
NON CALC

SST: OFFWH-LTBRN, TRNSL
F-VFGRN, RND-WRND
WSRTD, FRI, DUL YEL FLR
MOD BLOOM MKY-WH
CUT FLR, NO VIS CUT COL

SH: GY- DKGY, MOHD-HD
BLKY-PLTY, SUBFISS
NON CALC

SST: OFFWH-LTBRN, TRNSL
F-VFGRN, RND-WRND
WSRTD, FRI, ARG, DUL YEL
FLR, MOD BLOOM MKY-WH
CUT FLR, NO VIS CUT COL

SST: GY-LTGY, F-VFG
SUBRND-RND, WSRTD, FRI-
CONS, ARG, DUL YEL FLR, F
STRM CUT FLOR, NO VIS C

SPP:2700-2900
RPM:110-120
WOB:5-15
TORQ:17-20

MD:10957.54
INCL:49.01
AZM:178.95
TVD:10496.42

SPP:2700-2800psi
RPM:110-120
TORQ:19-22
WOB: 5-10

10/03/2014 MND=11480ft
11/03/2014

MD:11532.89'
INCL: 46.79
AZM:179.67
TVD:10812.87

MAX GAS=52units
C1=6268.8
C2=2194.08
C3=1149.28
C4=626.88

MAX GAS=47units
C1=6807.02
C2=1727.21
C3=679.76

MAX GAS: 48units
C1= 6345.3
C2= 2050.02
C3= 1269.06

MW:10.2 ppg
PV:28
YP:40
FV:80

MW:10.2
PV:28
YP:40
FV:80

SST:GY-LTBRN,F-VFGRN
SUBRND-RND,WSRTD
DUL YEL FLR,FST ST
MKY WH CUT FLOR,NO VIS
CUT FLR

SST:GY-LTGY,F-VFGRN
SUBRND-RND,WSRTD
DUL YEL FLR
NO VIS CUT FLR

SH:GY- DKGY,MOHD-HD
BLKY-PLTY,SUBFISS
SL CALC,SLTY

SH:GY- DKGY,MOHD-HD
BLKY-PLTY,SUBFISS
SL CALC,SLTY

SST:GY-LTGY,F-VFGRN
SUBRND-RND,WSRTD
FRIA,DUL YEL FLR,FST
STRM CUT FLR

SH:DKGY,MOHD-HD
BLKY-PLTY,SUBFISS-FISS
SLY CALC

SH:DKGY,MOHD-HD
BLKY-PLTY,SUBFISS-FISS
SLY CALC

SST:GY-LTGY,F-VFGRN
RND-WRND,WSRTD
FRIA,PINPOINT YEL FLR,
SLOW STRM MLKY WHT FLR
NO VIS CUT COLOR

SST:GY-LTGY,F-VFGRN
RND-WRND,WSRTD
FRIA,PINPOINT YEL FLR,
SLOW STRM MLKY WHT FLR
NO VIS CUT COLOR

SPP:2700-2950psi
RPM:110-120
TORQ:19-22
WOB: 5-10

MD:12015.22
INCL:38.01
AZM:179.97
TVD:11156.99

11/03/2014 MND=12056ft
12/03/2014

MD:12397.51
INCL:26.46
AZM:184.24
TVD:11484.85

SPP:2800-3000psi
RPM:110-125
TORQ:19-24
WOB: 5-10

12/03/2014 MND=12662ft
13/03/2014

11900

12000

12100

12200

12300

12400

12500

12600

12700

MAX: GAS: 17units
C1= 3457.19

MAX: GAS: 22units
C1= 4352.35

MAX: GAS: 23units
C1= 4618.99

MW:10.2
PV:28
YP:38
FV:75

MW:10.2
PV:28
YP:38
FV:75

SH:DKGY,MOHD-HD
BLKY-PLTY,SUBFISS-FISS
SLY CALC

SH:DKGY,HD,
BLKY-PLTY,FISS
SLY CALC

SH:DKGY,HD,
BLKY-PLTY,FISS
SLY CALC

SH:DKGY,HD,
BLKY-PLTY,FISS
SLY CALC

SH:DKGY,HD,
BLKY-PLTY,FISS
SLY SNDY

SH:DKGY,HD,
BLKY-PLTY,FISS
SLY SNDY,SLY CALC

